Quantification: 543_3
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	12
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2350.2	2188.5	668.925	px^2
major_axis_length_px	lacuna shape	91.5632	84.5141	33.0603	px
minor_axis_length_px	lacuna shape	36.5948	35.0553	10.2044	px
aspect_ratio	lacuna shape	2.6601	2.3867	1.276	unitless
eccentricity	lacuna shape	0.889	0.9075	0.078	unitless
solidity	lacuna shape	0.8742	0.9204	0.1013	unitless
roots_count	headline	8.1667	7	4.324	count
ring_length_r30_px	headline	411.417	414	106.432	px
ring_length_r60_px	reported	1076.25	1064	208.783	px
owned_length_px	ownership-dependent	3636.52	3773.94	1147.36	px
edge_count	ownership-dependent	168.75	163	61.2983	count
mean_edge_length_px	ownership-dependent	21.9749	22.0258	1.7723	px
perimeter_px	normalised	221.015	219.693	100.917	px
ring_area_r30_px2	normalised	8668.5	8576.5	2568.76	px^2
ring_area_r60_px2	normalised	21815.7	21875	4394.01	px^2
in_frame_fraction_r30	normalised	0.9974	1	0.009	unitless
in_frame_fraction_r60	normalised	0.9801	1	0.058	unitless
ring_density_r30	normalised	0.0491524	0.0465214	0.0115897	px^-1
ring_density_r60	normalised	0.0500524	0.0485338	0.00781146	px^-1
roots_per_100px_perimeter	normalised	4.0915	3.569	2.1734	per 100 px
ring_attached_length_r30_px	network v2	384.917	385.5	111.703	px
ring_attached_length_r60_px	network v2	970.083	989.5	235.169	px
ring_length_w_r30_px	network v2	467.79	465.019	121.117	px
ring_length_w_r60_px	network v2	1228.26	1208.44	236.062	px
sholl_crossings_r10	network v2	10.9167	10	3.9418	count
sholl_crossings_r20	network v2	13.4167	14	2.6785	count
sholl_crossings_r30	network v2	15.5	15.5	3.4245	count
loop_count	loops, report only	19.5	20.5	11.9507	count
loop_edge_share	loops, report only	0.4175	0.4053	0.1426	share
ring_width_mean_r30_px	width	7.1769	7.4567	0.9937	px

Per field
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measure	value	units
analysed_area_px2	1.02406e+06	px^2
total_skeleton_length_px	44604	px
canalicular_length_density_per_px	0.043556	px^-1
junction_count	1171	count
junction_density_per_px2	0.00114349	px^-2
lacuna_count	13	count

lacunae_per_px2	1.269e-05	px^-2
skeleton_component_count	142	count
mean_component_length_px	314.113	px
median_component_length_px	18.5	px
field_length_density_w_per_px	0.049656	px^-1
field_density_without_flagged_per_px	0.043556	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	211	px
unattached_component_count	14	count
canalicular_length_density_attached_per_px	0.043556	px^-1
field_length_density_with_unattached_per_px	0.043762	px^-1
loop_count	286	count
loop_edge_share	0.4916	share
edge_count	2317	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.9003	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	44604	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2174	80.8669	35.2412	2.2947	0.9	0.9518	0.4109	79.3965	270.898	False
2	3447	90.3677	63.9486	1.4131	0.7066	0.674	0.4223	197.059	916.902	False
3	1883	69.2617	34.8694	1.9863	0.864	0.9448	0.5742	250.833	340.47	False
4	3536	181.867	30.1414	6.0338	0.9862	0.7148	0.0738	327.046	42.4075	False
5	2052	69.2015	39.4228	1.7554	0.8219	0.9112	0.0622	388.671	369.939	False
6	2266	88.1613	35.5691	2.4786	0.915	0.9021	0.1324	473.05	745.191	False
7	1625	80.1248	27.6985	2.8927	0.9383	0.8325	0.2321	545.572	877.076	False
8	1683	71.0123	31.3735	2.2634	0.8971	0.9355	-0.0518	634.68	198.315	False
9	2203	91.2092	31.4053	2.9043	0.9389	0.9295	0.0832	734.301	635.44	False
10	2633	93.5592	36.2778	2.579	0.9218	0.9454	0.0457	780.576	481.495	False
11	531	28.7881	24.8252	1.1596	0.5063	0.9332	0.5942	1012.28	184.834	True
12								458.12	481.03	False
13								857.425	285.616	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	414	1151	2171.94	96	22.6243	6.8583
2	14	603	1467	4465.98	214	20.8691	7.0779
3	8	414	1065	4562.12	207	22.0392	8.3894
4	18	543	1063	3915.32	199	19.675	7.6996
5	7	376	970	3345.89	152	22.0124	7.7264
6	12	425	1157	2473.82	109	22.6956	8.1921

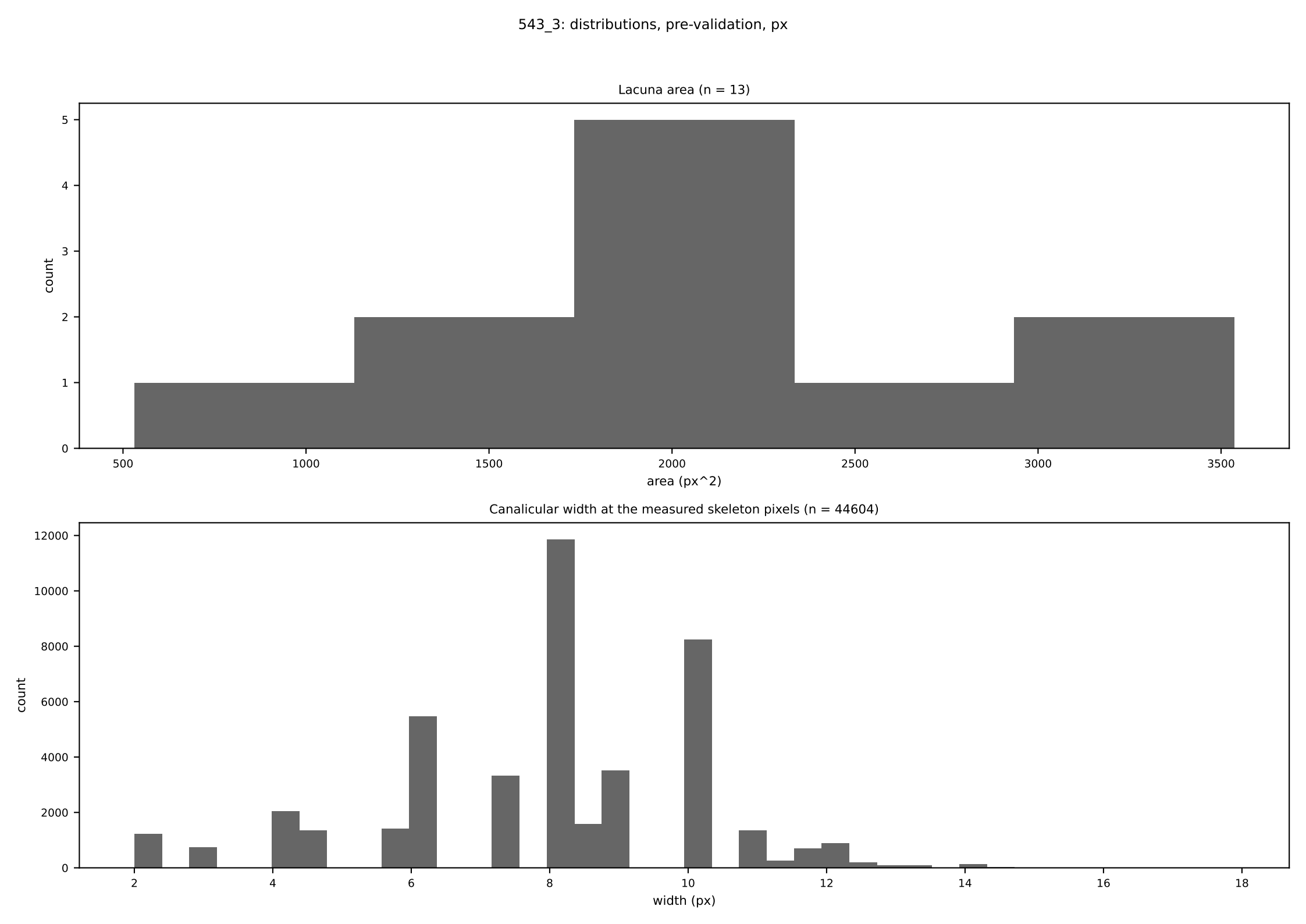
7	6	523	1388	5730.31	293	19.5574	6.3897
8	5	320	1027	4546.21	212	21.4444	7.6072
9	6	432	1139	4022.11	165	24.3764	7.7086
10	7	330	905	1757.89	69	25.4767	7.3063
11	2	193	577	1477.47	75	19.6996	6.3485
12	5	333	855	3632.55	161	22.5624	4.7196
13	3	224	728	3014.08	148	20.3654	6.4471

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	210.066	8595	21715	1	0.9616	0.0481675	0.0530048	3.3323
2	363.446	12097	29071	1	1	0.0498471	0.0504627	3.852
3	184.794	7941	19515	1	1	0.0521345	0.0545734	4.3291
4	415.99	13536	26753	0.9687	0.7992	0.0401152	0.0397339	4.327
5	205.965	8333	20013	1	1	0.0451218	0.0484685	3.3986
6	241.907	9380	23807	1	1	0.0453092	0.0485992	4.9606
7	229.321	8558	22035	1	1	0.0611124	0.0629907	2.6164
8	191.965	7943	21525	1	1	0.040287	0.047712	2.6046
9	238.108	9212	24077	1	1	0.0468954	0.0473066	2.5199
10	242.936	9386	24420	1	1	0.0351588	0.0370598	2.8814
11	95.3909	3354	9525	0.6102	0.5725	0.0575432	0.0605774	2.0966
12	47.4558	4187	13513	1	1	0.0795319	0.0632724	10.5361
13	80.2254	4854	15344	1	1	0.0461475	0.0474453	3.7395

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	379	981	470.718	1335.28	9	15	18
2	578	1374	696.955	1672.82	19	16	19
3	392	1049	459.321	1211.83	10	13	18
4	543	1010	608.174	1205.05	17	16	10
5	365	903	420.635	1116.52	10	15	14
6	425	998	477.777	1311.95	11	14	14
7	466	1321	602.801	1582.25	12	16	22
8	289	936	367.392	1169.28	9	10	15
9	413	1051	489.676	1281.93	13	14	16
10	257	648	374.291	1022.26	9	14	13
11	153	333	220.51	657.458	4	5	9
12	314	817	382.877	996.134	7	10	16
13	198	553	262.865	833.796	5	8	11


543_3: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 44604)
count
width (px)